

TERROR FINANCING, MONEY LAUNDERING AND FATF

FINANCE-PURPOSE MATRIX → PREDICATE-PROCEEDS GATEWAY → LAUNDERING-STAGE TYPOLOGY → CHANNEL DISTINCTION MAP → PREVENTIVE-CONTROL CHAIN → PMLA INSTITUTION LADDER

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g2 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00

FINANCE-PURPOSE MATRIX

FINANCE-PURPOSE MATRIX MONEY LAUNDERING DISGUISE PROCEEDS OF CRIME TERROR FINANCING MOVE VALUE FOR TERROR PURPOSE MAY BE LAWFUL OR UNLAWFUL IN ORIGIN PURPOSE, SOURCE AND LEGAL ROUTE MUST ALL BE NAMED

THREAT / INSTITUTION / INSTRUMENT	DECISIVE LEGAL OR STATUS DISTINCTION	ATTRIBUTION / IMPLEMENTATION LIMIT
MONEY LAUNDERING → DISGUISE PROCEEDS OF CRIME	FUNDS → MAY BE LAWFUL OR UNLAWFUL IN ORIGIN	MONEY LAUNDERING → DISGUISE PROCEEDS OF CRIME
TERROR FINANCING → PROVIDE / MOVE VALUE FOR TERROR PURPOSE	RULE → purpose, source and legal route must all be named	TERROR FINANCING → PROVIDE / MOVE VALUE FOR TERROR PURPOSE
THREAT / INSTITUTION / INSTRUMENT - MONEY LAUNDERING → DISGUISE PROCEEDS OF CRIME - TERROR FINANCING → PROVIDE / MOVE VALUE FOR TERROR PURPOSE	DECISIVE LEGAL OR STATUS DISTINCTION - FUNDS → MAY BE LAWFUL OR UNLAWFUL IN ORIGIN - RULE → purpose, source and legal route must all be named	ATTRIBUTION / IMPLEMENTATION LIMIT - MONEY LAUNDERING → DISGUISE PROCEEDS OF CRIME - TERROR FINANCING → PROVIDE / MOVE VALUE FOR TERROR PURPOSE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → purpose, source and legal route must all be named.

01

STAGE 01

PREDICATE-PROCEEDS GATEWAY

PREDICATE-PROCEEDS GATEWAY SCHEDULED PREDICATE OFFENCE REGISTERED PROCEEDS OF CRIME IDENTIFIED PMLA INVESTIGATION UNEXPLAINED WEALTH ALONE IS NOT THE COMPLETE GATEWAY

SCHEDULED PREDICATE OFFENCE REGISTERED → PROCEEDS OF CRIME IDENTIFIED → PMLA INVESTIGATION → PROSECUTION / ADJUDICATION → TRAP → unexplained wealth alone is not the complete gateway

STARTING CONDITION / THREAT	GOVERNANCE / RESPONSE MECHANISM	RIGHTS / STATUS / OUTCOME LIMIT
- SCHEDULED PREDICATE OFFENCE REGISTERED - PROCEEDS OF CRIME IDENTIFIED	- PMLA INVESTIGATION - PROSECUTION / ADJUDICATION	- TRAP → unexplained wealth alone is not the complete gateway - SCHEDULED PREDICATE OFFENCE REGISTERED

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → unexplained wealth alone is not the complete gateway.

02

STAGE 02

LAUNDERING-STAGE TYPOLOGY

LAUNDERING-STAGE TYPOLOGY VALUE ENTERS A CHANNEL DISTANCE FROM CRIMINAL ORIGIN APPARENTLY LEGITIMATE RETURN SAFE RULE CONCEPTUAL DIAGNOSIS, NEVER EVASION INSTRUCTION

PLACEMENT → VALUE ENTERS A CHANNEL → LAYERING → DISTANCE FROM CRIMINAL ORIGIN → INTEGRATION → APPARENTLY LEGITIMATE RETURN → SAFE RULE → conceptual diagnosis, never evasion instruction

STARTING CONDITION / THREAT	GOVERNANCE / RESPONSE MECHANISM	RIGHTS / STATUS / OUTCOME LIMIT
- PLACEMENT → VALUE ENTERS A CHANNEL - LAYERING → DISTANCE FROM CRIMINAL ORIGIN	- INTEGRATION → APPARENTLY LEGITIMATE RETURN - SAFE RULE → conceptual diagnosis, never evasion instruction	- PLACEMENT → VALUE ENTERS A CHANNEL - LAYERING → DISTANCE FROM CRIMINAL ORIGIN

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SAFE RULE → conceptual diagnosis, never evasion instruction.

03

STAGE 03

CHANNEL DISTINCTION MAP

CHANNEL DISTINCTION MAP TRADE VALUE DOCUMENT MANIPULATION BANKING CHANNEL TRANSACTION CONTROLS VIRTUAL ASSET SEPARATE TECHNOLOGY REPORTING RISK

THREAT / INSTITUTION / INSTRUMENT	DECISIVE LEGAL OR STATUS DISTINCTION	ATTRIBUTION / IMPLEMENTATION LIMIT
TBML → TRADE VALUE / DOCUMENT MANIPULATION	VIRTUAL ASSET → SEPARATE TECHNOLOGY / REPORTING RISK	TBML → TRADE VALUE / DOCUMENT MANIPULATION
BANKING CHANNEL → REPORTING / TRANSACTION CONTROLS	TERROR DIVERSION → MAY START WITH LAWFUL FUNDS	BANKING CHANNEL → REPORTING / TRANSACTION CONTROLS
THREAT / INSTITUTION / INSTRUMENT - TBML → TRADE VALUE / DOCUMENT MANIPULATION - BANKING CHANNEL → REPORTING / TRANSACTION CONTROLS	DECISIVE LEGAL OR STATUS DISTINCTION - VIRTUAL ASSET → SEPARATE TECHNOLOGY / REPORTING RISK - TERROR DIVERSION → MAY START WITH LAWFUL FUNDS	ATTRIBUTION / IMPLEMENTATION LIMIT - TBML → TRADE VALUE / DOCUMENT MANIPULATION - BANKING CHANNEL → REPORTING / TRANSACTION CONTROLS

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SAFE RULE — conceptual diagnosis, never evasion instruction.

03

STAGE 03 CHANNEL DISTINCTION MAP

CHANNEL DISTINCTION MAP TRADE VALUE DOCUMENT MANIPULATION BANKING CHANNEL TRANSACTION CONTROLS VIRTUAL ASSET SEPARATE TECHNOLOGY REPORTING RISK

THREAT / INSTITUTION / INSTRUMENT

TBML → TRADE VALUE / DOCUMENT MANIPULATION

BANKING CHANNEL → REPORTING / TRANSACTION CONTROLS

THREAT / INSTITUTION / INSTRUMENT

• TBML → TRADE VALUE / DOCUMENT MANIPULATION

• BANKING CHANNEL → REPORTING / TRANSACTION CONTROLS

DECISIVE LEGAL OR STATUS DISTINCTION

VIRTUAL ASSET → SEPARATE TECHNOLOGY / REPORTING RISK

TERROR DIVERSION → MAY START WITH LAWFUL FUNDS

DECISIVE LEGAL OR STATUS DISTINCTION

• VIRTUAL ASSET → SEPARATE TECHNOLOGY / REPORTING RISK

• TERROR DIVERSION → MAY START WITH LAWFUL FUNDS

ATTRIBUTION / IMPLEMENTATION LIMIT

TBML → TRADE VALUE / DOCUMENT MANIPULATION

BANKING CHANNEL → REPORTING / TRANSACTION CONTROLS

ATTRIBUTION / IMPLEMENTATION LIMIT

• TBML → TRADE VALUE / DOCUMENT MANIPULATION

• BANKING CHANNEL → REPORTING / TRANSACTION CONTROLS

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TERROR DIVERSION — MAY START WITH LAWFUL FUNDS.

04

STAGE 04 PREVENTIVE-CONTROL CHAIN

PREVENTIVE-CONTROL CHAIN BENEFICIAL OWNER TRANSACTION MONITORING SUSPICIOUS TRANSACTION REPORT FIU-IND ANALYSIS NOT PROVED CRIME OR GUILT

CDD / KYC → BENEFICIAL OWNER

RECORDS / TRANSACTION MONITORING

SUSPICIOUS TRANSACTION REPORT

FIU-IND ANALYSIS / DISSEMINATION

NOT PROVED — crime or guilt

STARTING CONDITION / THREAT

• CDD / KYC → BENEFICIAL OWNER

• RECORDS / TRANSACTION MONITORING

GOVERNANCE / RESPONSE MECHANISM

• SUSPICIOUS TRANSACTION REPORT

• FIU-IND ANALYSIS / DISSEMINATION

RIGHTS / STATUS / OUTCOME LIMIT

• NOT PROVED → crime or guilt

• CDD / KYC → BENEFICIAL OWNER

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FIU-IND ANALYSIS / DISSEMINATION; NOT PROVED — crime or guilt.

05

STAGE 05 PMLA INSTITUTION LADDER

PMLA INSTITUTION LADDER PREDICATE AGENCY UNDERLYING OFFENCE FINANCIAL INTELLIGENCE PMLA INVESTIGATION PROVISIONAL ATTACHMENT ADJUDICATING AUTHORITY SPECIAL COURT

PREDICATE AGENCY → UNDERLYING OFFENCE

FIU-IND → FINANCIAL INTELLIGENCE

ED → PMLA INVESTIGATION / PROVISIONAL ATTACHMENT

ADJUDICATING AUTHORITY → CONFIRM / RELEASE

STARTING CONDITION / THREAT

• PREDICATE AGENCY → UNDERLYING OFFENCE

• FIU-IND → FINANCIAL INTELLIGENCE

GOVERNANCE / RESPONSE MECHANISM

• ED → PMLA INVESTIGATION / PROVISIONAL ATTACHMENT

• ADJUDICATING AUTHORITY → CONFIRM / RELEASE

RIGHTS / STATUS / OUTCOME LIMIT

• SPECIAL COURT → TRIAL / LEGAL OUTCOME

• PREDICATE AGENCY → UNDERLYING OFFENCE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ADJUDICATING AUTHORITY → CONFIRM / RELEASE; SPECIAL COURT → TRIAL / LEGAL OUTCOME.

06

STAGE 06 PROPERTY-STATUS FIREWALL

PROPERTY-STATUS FIREWALL PROVISIONAL ATTACHMENT CONFIRMATION OR RELEASE CONVICTION OR ACQUITTAL CONFISCATION ONLY ON THE LEGALLY REQUIRED OUTCOME

TRACE / IDENTIFY

PROVISIONAL ATTACHMENT

CONFIRMATION OR RELEASE

CONVICTION OR ACQUITTAL

THREAT / INSTITUTION / INSTRUMENT

• TRACE / IDENTIFY

• PROVISIONAL ATTACHMENT

DECISIVE LEGAL OR STATUS DISTINCTION

• CONFIRMATION OR RELEASE

• CONVICTION OR ACQUITTAL

ATTRIBUTION / IMPLEMENTATION LIMIT

• CONFISCATION ONLY ON THE LEGALLY REQUIRED OUTCOME

• TRACE / IDENTIFY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CONVICTION OR ACQUITTAL; CONFISCATION ONLY ON THE LEGALLY REQUIRED OUTCOME.

06

STAGE 06 PROPERTY-STATUS FIREWALL

PROPERTY-STATUS FIREWALL PROVISIONAL ATTACHMENT CONFIRMATION OR RELEASE CONVICTION OR ACQUITTAL CONFISCATION ONLY ON THE LEGALLY REQUIRED OUTCOME

TRACE / IDENTIFY

PROVISIONAL ATTACHMENT

CONFIRMATION OR RELEASE

CONVICTION OR ACQUITTAL

THREAT / INSTITUTION / INSTRUMENT

- TRACE / IDENTIFY
- PROVISIONAL ATTACHMENT

DECISIVE LEGAL OR STATUS DISTINCTION

- CONFIRMATION OR RELEASE
- CONVICTION OR ACQUITTAL

ATTRIBUTION / IMPLEMENTATION LIMIT

- CONFISCATION ONLY ON THE LEGALLY REQUIRED OUTCOME
- TRACE / IDENTIFY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CONVICTION OR ACQUITTAL; CONFISCATION ONLY ON THE LEGALLY REQUIRED OUTCOME.

07

STAGE 07 TERROR-SANCTIONS ROUTE

TERROR-SANCTIONS ROUTE TERROR-FUND OFFENCE? UAPA INVESTIGATION LISTED PERSON SECTION 51A TARGETED ACTION UN 1267 LINK DOMESTIC IMPLEMENTATION IDENTITY ACCURACY

TERROR-FUND OFFENCE? → UAPA INVESTIGATION / TRIAL

LISTED PERSON / ENTITY? → SECTION 51A TARGETED ACTION

UN 1267 LINK → DOMESTIC IMPLEMENTATION

SAFEGUARD → REVIEW / DELISTING / IDENTITY ACCURACY

THREAT / GOVERNANCE CONCEPT

- TERROR-FUND OFFENCE? → UAPA INVESTIGATION / TRIAL
- LISTED PERSON / ENTITY? → SECTION 51A TARGETED ACTION

INSTITUTIONAL RESPONSE

- UN 1267 LINK → DOMESTIC IMPLEMENTATION
- SAFEGUARD → REVIEW / DELISTING / IDENTITY ACCURACY

LEGAL / EVIDENTIARY LIMIT

- TERROR-FUND OFFENCE? → UAPA INVESTIGATION / TRIAL
- LISTED PERSON / ENTITY? → SECTION 51A TARGETED ACTION

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SAFEGUARD → REVIEW / DELISTING / IDENTITY ACCURACY.

08

STAGE 08 THREE-FINANCE-LIMBS

PROCEEDS OF CRIME FUNDS FOR TERROR PURPOSE PROHIBITED WMD-RELATED FINANCING COMMON NEED RISK CONTROLS + INTELLIGENCE + LAWFUL ADJUDICATION

THREAT / INSTITUTION / INSTRUMENT

AML → PROCEEDS OF CRIME
CFT → FUNDS FOR TERROR PURPOSE

DECISIVE LEGAL OR STATUS DISTINCTION

PF → PROHIBITED WMD-RELATED FINANCING
COMMON NEED → risk controls + intelligence + lawful adjudication

ATTRIBUTION / IMPLEMENTATION LIMIT

AML → PROCEEDS OF CRIME
CFT → FUNDS FOR TERROR PURPOSE

THREAT / INSTITUTION / INSTRUMENT

- AML → PROCEEDS OF CRIME
- CFT → FUNDS FOR TERROR PURPOSE

DECISIVE LEGAL OR STATUS DISTINCTION

- PF → PROHIBITED WMD-RELATED FINANCING
- COMMON NEED → risk controls + intelligence + lawful adjudication

ATTRIBUTION / IMPLEMENTATION LIMIT

- AML → PROCEEDS OF CRIME
- CFT → FUNDS FOR TERROR PURPOSE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: COMMON NEED → risk controls + intelligence + lawful adjudication.

09

STAGE 09 FATF ASSESSMENT ARCHITECTURE

FATF ASSESSMENT ARCHITECTURE FATF RECOMMENDATIONS INTERNATIONAL STANDARDS MUTUAL EVALUATION TECHNICAL COMPLIANCE + EFFECTIVENESS POST-EVALUATION REPORTING INCREASED MONITORING CALL FOR ACTION

FATF RECOMMENDATIONS → INTERNATIONAL STANDARDS

MUTUAL EVALUATION → TECHNICAL COMPLIANCE + EFFECTIVENESS

FOLLOW-UP → POST-EVALUATION REPORTING

LISTING → INCREASED MONITORING / CALL FOR ACTION

THREAT / GOVERNANCE CONCEPT

- FATF RECOMMENDATIONS → INTERNATIONAL STANDARDS
- MUTUAL EVALUATION → TECHNICAL COMPLIANCE + EFFECTIVENESS

INSTITUTIONAL RESPONSE

- FOLLOW-UP → POST-EVALUATION REPORTING
- LISTING → INCREASED MONITORING / CALL FOR ACTION

LEGAL / EVIDENTIARY LIMIT

- MUST REMEMBER: Terror finance and money laundering analysis follows predicate offence,...
- FATF RECOMMENDATIONS → INTERNATIONAL STANDARDS

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Terror finance and money laundering analysis follows predicate offence,.

09

STAGE 09 FATF ASSESSMENT ARCHITECTURE

FATF ASSESSMENT ARCHITECTURE FATF RECOMMENDATIONS INTERNATIONAL STANDARDS MUTUAL EVALUATION TECHNICAL COMPLIANCE + EFFECTIVENESS POST-EVALUATION REPORTING INCREASED MONITORING CALL FOR ACTION

FATF RECOMMENDATIONS → INTERNATIONAL STANDARDS

MUTUAL EVALUATION → TECHNICAL COMPLIANCE + EFFECTIVENESS

FOLLOW-UP → POST-EVALUATION REPORTING

LISTING → INCREASED MONITORING / CALL FOR ACTION

THREAT / GOVERNANCE CONCEPT

- FATF RECOMMENDATIONS → INTERNATIONAL STANDARDS
- MUTUAL EVALUATION → TECHNICAL COMPLIANCE + EFFECTIVENESS

INSTITUTIONAL RESPONSE

- FOLLOW-UP → POST-EVALUATION REPORTING
- LISTING → INCREASED MONITORING / CALL FOR ACTION

LEGAL / EVIDENTIARY LIMIT

- MUST REMEMBER: Terror finance and money laundering analysis follows predicate offence,...
- FATF RECOMMENDATIONS → INTERNATIONAL STANDARDS

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Terror finance and money laundering analysis follows predicate offence,.

10

STAGE 10 INDIA 2024 BALANCED SCORECARD

INDIA 2024 BALANCED SCORECARD RISK UNDERSTANDING FINANCIAL INTELLIGENCE BENEFICIAL OWNERSHIP ASSET RECOVERY COMPLETE ML TF TRIALS + APPROPRIATE SANCTIONS RISK-BASED NON-PROFIT OUTREACH

THREAT / INSTITUTION / INSTRUMENT

STRENGTH → RISK UNDERSTANDING / FINANCIAL INTELLIGENCE
STRENGTH → BENEFICIAL OWNERSHIP / ASSET RECOVERY

DECISIVE LEGAL OR STATUS DISTINCTION

GAP → COMPLETE ML / TF TRIALS + APPROPRIATE SANCTIONS
GAP → RISK-BASED NON-PROFIT OUTREACH

ATTRIBUTION / IMPLEMENTATION LIMIT

STATUS → REGULAR FOLLOW-UP; REPORT BACK IN THREE YEARS
CLOSE DISTINCTION: FATF standard is not Indian conviction, grey listing is not sanction,...

THREAT / INSTITUTION / INSTRUMENT

- STRENGTH → RISK UNDERSTANDING / FINANCIAL INTELLIGENCE
- STRENGTH → BENEFICIAL OWNERSHIP / ASSET RECOVERY

DECISIVE LEGAL OR STATUS DISTINCTION

- GAP → COMPLETE ML / TF TRIALS + APPROPRIATE SANCTIONS
- GAP → RISK-BASED NON-PROFIT OUTREACH

ATTRIBUTION / IMPLEMENTATION LIMIT

- STATUS → REGULAR FOLLOW-UP; REPORT BACK IN THREE YEARS
- CLOSE DISTINCTION: FATF standard is not Indian conviction, grey listing is not sanction,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: FATF standard is not Indian conviction, grey listing is not sanction,.

11

STAGE 11 PYQ AND EFFECTIVENESS RAIL

PYQ AND EFFECTIVENESS RAIL GLOBALISATION + MATCHED CONTROLS FUNDING SOURCES + FATF COMPLIANCE EFFORTS CROSS-OWNED DRUG LAUNDERING LINK ATTACHMENT, LISTING AND COMPLIANCE ARE NOT CONVICTION EVIDENCE LIMIT USE PMLA, FIU-IND, ED, NIA AND FATF WITH

2021 → TECHNOLOGY / GLOBALISATION + MATCHED CONTROLS

2023 → FUNDING SOURCES + FATF COMPLIANCE EFFORTS

2018 → CROSS-OWNED DRUG / LAUNDERING LINK

END → PREVENT / DETECT / DISRUPT / PROSECUTE / ADJUDICATE

QUALIFY → attachment, listing and compliance are not conviction

THREAT / GOVERNANCE CONCEPT

- 2021 → TECHNOLOGY / GLOBALISATION + MATCHED CONTROLS
- 2023 → FUNDING SOURCES + FATF COMPLIANCE EFFORTS

INSTITUTIONAL RESPONSE

- 2018 → CROSS-OWNED DRUG / LAUNDERING LINK
- END → PREVENT / DETECT / DISRUPT / PROSECUTE / ADJUDICATE

LEGAL / EVIDENTIARY LIMIT

- QUALIFY → attachment, listing and compliance are not conviction
- EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Use PMLA, FIU-IND, ED, NIA and FATF with...

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Use PMLA, FIU-IND, ED, NIA and FATF with...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Use PMLA, FIU-IND, ED, NIA and FATF with.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT RECEIVES AND ANALYSES SUSPICIOUS TRANSACTION REPORTS/CASH ENFORCEMENT DIRECTORATE (ED) INVESTIGATES PMLA CASES TIED TO A SPECIAL COURTS UNDER PMLA THE PROSECUTION LAYER FATF'S 2024

THE PREDICATE-OFFENCE DEPENDENCY AS AN INSTITUTIONAL DESIGN UAPA SECTION 51A AND THE UN 1267 ROUTE

OPTIONAL DOCTRINE / INSTITUTION DEPTH

- FIU-IND: receives and analyses Suspicious Transaction Reports/Cash
- Enforcement Directorate (ED): investigates PMLA cases tied to a
- CAUTION: Special Courts under PMLA: the prosecution layer FATF's 2024
- CAUTION: The predicate-offence dependency as an institutional design

LAW, ATTRIBUTION OR STATUS LIMIT

- UAPA Section 51A and the UN 1267 route: the terrorist-financing
- FATF's three-year progress-report obligation (in 2027): the
- CAUTION: Emerging virtual-asset regulatory architecture: CAUTION: any specific
- CAUTION: The 26-country MLAT figure (Singh, book-period) as a boundary case

COMPARATIVE GOVERNANCE / RIGHTS NUANCE

- CAUTION: JAM-Trinity (Jan Dhan-Aadhaar-Mobile) as an AML/CFT-relevant boundary
- CAUTION: Non-profit-sector terror-financing risk as a boundary case: FATF's
- CAUTION: Strong detection/seizure capacity without matching prosecutorial
- CAUTION: TBML detection depends on customs-agency focus shifting from

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.